

Basic Network Sniffer Using Python and Scapy

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Abstract

The objective of this project is to develop a basic network sniffer using Python and the Scapy library. The application captures network packets in real time and analyzes source IP addresses, destination IP addresses, protocols, ports, and payload data.

Introduction

Network sniffing is the process of capturing and analyzing packets moving across a network. Packet sniffers are used for monitoring, troubleshooting, and understanding network communication.

Objectives

Capture live network packets, analyze packet structure, identify protocols, display source and destination IP addresses, display port numbers and payloads, and understand network traffic flow.

Tools and Technologies

Python, Scapy, Windows or Kali Linux, TCP/IP protocols.

Methodology

The program uses Scapy's `sniff()` function to capture packets. Each packet is analyzed to determine protocol type, IP details, ports, and payload content.

Code Explanation

The application captures packets using `sniff()`, checks for TCP, UDP, and ICMP layers, extracts packet information, and displays payload data when available.

Results

The sniffer successfully captured and analyzed network traffic, displaying source and destination addresses, port information, protocol types, and payload data.

Advantages

Real-time monitoring, simple implementation, protocol analysis, and cybersecurity learning.

Limitations

Requires administrator privileges and cannot fully inspect encrypted HTTPS payloads.

Conclusion

The project successfully demonstrated packet capturing and protocol analysis using Python and Scapy, providing practical exposure to network security concepts.

References

Scapy Documentation, Python Documentation, TCP/IP Networking Fundamentals.